

Data Engineering and Machine Learning Pipeline

Phase II Project Report

ENGE707 Data Engineering and AI

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1. Introduction – Project Phase-I Summary

- ☐ Students must briefly summarise the Phase I project. Students are required to:
 - ☒ ~~Restate the selected problem, dataset, target variable, and prediction objective.~~
 - ☒ ~~Summarise the dataset source, number of records, number of features, and main data characteristics.~~
 - ☐ Briefly describe the main data-quality issues identified in Phase I.
 - ☐ Explain why the problem is suitable for an LCS-based solution.

Link with justification, do not repeat

- raw data, no preprocessing
- base elcs model
- apply preprocessing, we want to see if our preprocessing was good or not
 - Evidence between pre processing and the processed data
 - reducing the search space -
- Make some changes in the ELCS base system to improve the accuracy or computational cost
e.g. lowering the training amount (Novelty)
 - changing fitness function is novelty
- Training/Testing sets should be run at least 10 times. take the average of 10 times
 - use different seed
 - *also mention the seeds in the report.*
- cross fold validation not random and it is in sequence
- **don't forget to randomize when separating testing and training data**

1.1 Problem Restatement

In this project, the dataset “281K Patient Dataset for Multi-Disease Prediction” from Kaggle is used. The goal is to find whether the lifestyle, demographics, and clinical measurements in the dataset have any relevance to predicting whether an individual has heart disease. A model with this prediction ability isn’t a full replacement for having an in-person cardiovascular risk assessment. Still, it would be a cost-effective way to check for an individual’s chance of having heart disease. The target column “heart_disease_target” is created after cleaning and is based on the “sublabel” column, which contains the combination of diseases an instance has. If the instance from the sublabel contains “HT”, the value of the new column would be True; otherwise, False. This new column is created to be easily used to compare and train the machine learning model’s prediction capabilities.

1.2 Dataset summary

The dataset is a combination of three datasets (diabetes, heart disease, and hypertension). There are 280,985 raw records, and 278,701 remain after removing duplicates; the 278,701 will be used instead of the raw amount. The original dataset has 39 columns, but the introduction of “heart_disease_target” gives 40 columns. The columns utilized during the analysis and processing stage are lifestyle, demographics, clinical, and disease indicators. No missing values were found during the dataset inspection.

The overall heart disease rate is 2.1% across all these sources, but individually the dataset shows varying heart disease rates: 20.06% in the heart disease dataset source, compared to 4.17% for the diabetes source, and 0% for the hypertension source.

1.3 Data-quality issues identified in Phase I

From the data inspection and data cleaning stage, multiple issues in the dataset were identified:

- 2,284 duplicate instances were removed.

- heart_disease has fractional values (from 0 to 1), which doesn't clearly explain whether an individual has heart disease, so a new column (heart_disease_target) was created to serve as the target (binary) instead of the heart_disease column.
- The columns sublabel, disease_flags, label, and heart_disease all contain data leakage information. This provides clues and partial answers to whether the individual has heart disease; thus, they are excluded from the prediction process.
- Some columns such as age/age_level/age_normalized and bmi/bmi_level carry the same information; this is to be reviewed before modeling.
- Imbalance in the distribution of instances with and without heart disease, since 97.9% of records do not have heart disease and 2.1% do.
- As described in the dataset summary, individual sources have different heart disease rates.

1.4 Justification for an LCS-based solution

2. Original LCS System on Raw Dataset

- ☐ Students must run an original LCS system on the raw or original dataset before making improvements. Students are required to:
 - ☐ Select an appropriate LCS variant, preferably eLCS.
 - ☐ Use the original LCS code without algorithmic modification.
 - ☐ Run the original LCS system on the raw or minimally processed dataset.
 - ☐ Report the selected LCS parameters.
 - ☐ Record the baseline performance results.
 - ☐ Explain any minimal processing required to make the dataset compatible with the LCS code.

2.1 LCS variant selection (preferably eLCS) – justification

2.2 Confirmation that original LCS code is used without algorithmic modification

2.3 Execution on raw / minimally processed dataset

2.4 LCS parameter settings used

2.5 Baseline performance results

2.6 Minimal processing required for LCS compatibility

3. Data Preprocessing and Feature Engineering

- ☐ Students must improve the quality of the dataset before developing the improved system and make it suitable for working with LCS. Students are required to:
 - ☐ Address missing values, duplicate records, invalid values, inconsistent categories, and incorrect data types.
 - ☐ Investigate and handle outliers where appropriate.
 - ☐ Encode categorical variables and scale, discretise, or transform numerical variables where required.
 - ☐ Address class imbalance where relevant.
 - ☐ Apply feature engineering, feature selection, or dimensionality reduction where justified.
 - ☐ Clearly explain how preprocessing is expected to improve LCS performance.
 - ☐ Ensure that preprocessing avoids data leakage.

3.1 Data cleaning – missing values, duplicates, invalid values, inconsistent categories, incorrect data types

3.2 Outlier investigation and handling

3.3 Encoding categorical variables; scaling/discretising/transforming numerical variables

3.4 Class-imbalance handling

3.5 Feature engineering, feature selection, or dimensionality reduction – justification

3.6 Expected impact of preprocessing on LCS performance

3.7 Data-leakage prevention measures

4. Improved LCS-Based System

☐ Students must improve the LCS code, LCS configuration, or machine learning pipeline to develop a novel improved system. Possible improvements include:

- ☐ Hyperparameter tuning.
- ☐ Improved preprocessing designed for LCS performance.
- ☐ Feature selection or dimensionality reduction.
- ☐ Rule pruning or rule compaction.
- ☐ Modified LCS algorithm.
- ☐ Class-imbalance handling.
- ☐ Cost-sensitive learning.

☐ Students are required to:

- ☐ Clearly identify what was changed compared with the original LCS system.
- ☐ Justify why the change may improve prediction accuracy or model quality.

- ☐ Provide a brief algorithmic explanation, pseudocode, or flowchart of the improved system.
- ☐ Compare the improved system with the original LCS baseline.

4.1 Summary of changes vs. the original LCS system

4.2 Justification – expected effect on accuracy or model quality

4.3 Algorithmic explanation (pseudocode or flowchart)

4.4 Comparison against the original LCS baseline

5. Experiments

- ☐ Students must use a suitable and justified validation strategy. Students are required to:
 - ☐ Use either an 80/20 train-test split or k-fold cross-validation.
 - ☐ Justify the selected validation approach.
- ☐ Students must evaluate results using appropriate metrics and statistical tests. Students are required to:
 - ☐ Select evaluation metrics suitable for the selected problem.
 - ☐ For classification, consider accuracy, precision, recall, F1-score, balanced accuracy, ROC-AUC, PR-AUC, and confusion matrix.
 - ☐ For regression, consider MAE, MSE, RMSE, MAPE, R^2 , and residual analysis.
 - ☐ Justify the selected metrics.
 - ☐ Apply at least one suitable statistical test to compare model performance.
 - ☐ Interpret whether performance differences are statistically significant.

5.1 Validation strategy (80/20 split or k-fold CV) – justification

5.2 Evaluation metrics – selection and justification (classification: accuracy, precision, recall, F1, balanced accuracy, ROC-AUC, PR-AUC, confusion matrix; regression: MAE, MSE, RMSE, MAPE, R^2 , residual analysis)

5.3 Statistical significance testing

5.4 Interpretation of significance results

6. Model Comparison

- ☐ Students must compare the proposed improved LCS-based system with the original LCS and at least three conventional machine learning models. The comparison must include:
 - ☐ Original LCS on the raw or minimally processed dataset.
 - ☐ Original LCS on the preprocessed dataset.
 - ☐ Improved LCS-based system on the preprocessed and/or feature-engineered dataset.
 - ☐ At least three conventional non-deep-learning machine learning models.
- ☐ Conventional models may include Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, Naive Bayes, k-Nearest Neighbours, Gaussian Process, or other appropriate non-deep-learning models. Deep-learning models are not permitted for the required model comparison.

6.1 Original LCS – raw/minimally processed data

6.2 Original LCS – preprocessed data

6.3 Improved LCS – preprocessed/feature-engineered data

6.4 At least three conventional ML models (e.g. Random Forest, SVM, Decision Tree, Logistic Regression, Naive Bayes, k-NN, Gaussian Process) – no deep learning

6.5 Comparative results table

7. Interpretation of Results and Explainable AI

- ☐ Students must include a separate section on result interpretation and Explainable AI. Students are required to:
 - ☐ Interpret the main findings from the model comparison.
 - ☐ Explain important LCS rules, feature conditions, or classifier patterns.
 - ☐ Use at least three sample LCS rules/classifiers to discuss the decision-making process.
 - ☐ Discuss how LCS rule-based outputs support interpretability.
 - ☐ Discuss whether the explanations are trustworthy and practically meaningful.

7.1 Interpretation of main findings from the model comparison

7.2 Explanation of key LCS rules / feature conditions / classifier patterns

7.3 At least three sample LCS rules discussed for decision-making insight

7.4 How rule-based outputs support interpretability

7.5 Trustworthiness and practical meaningfulness of the explanations

8. Discussion, Limitations, and Future Work

☐ Students must critically discuss the results and the responsible use of the developed system. Students are required to discuss:

- ☐ Reasons for performance improvement or lack of improvement.
- ☐ Practical meaning of the results, not only numerical performance.
- ☐ Limitations due to data quality, dataset size, class imbalance, noise, missingness, or feature relevance.
- ☐ Future work, such as improved LCS variants, better preprocessing, external validation, or domain expert review.

8.1 Reasons for performance improvement (or lack thereof)

8.2 Practical significance of results beyond raw numbers

8.3 Limitations – data quality, dataset size, class imbalance, noise, missingness, feature relevance

8.4 Future work – improved LCS variants, better preprocessing, external validation, domain expert review

Notes

Final report: 5,000--6,500 words, excluding references and appendices.

Source code or Jupyter Notebook.

GitHub repository link and commit-history evidence.

Contribution statement, including a brief individual reflection from each group member.

Cleaned and feature-engineered dataset, or clear dataset reproduction instructions.

Turnitin similarity report with a similarity score not exceeding 20%.

References

Dataset

- <https://www.kaggle.com/datasets/danishjmeo/281k-patient-dataset-for-multi-disease-prediction?resource=download>

Sub Datasets

- <https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>,
- [kaggle.com/datasets/oktayrdeki/heart-disease](https://www.kaggle.com/datasets/oktayrdeki/heart-disease),
- [kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset](https://www.kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset)